The exact second moment of $\sum_{n<N} \Lambda(n)\mu(N-n)$, and why Chowla's conjecture does not control it

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Abstract

The Huang–Li reduction of binary Goldbach to Elliott–Halberstam [1], once its demand side is evaluated [4], rests on the single scalar

$$C(N) \;=\; \sum_{n<N} \Lambda(n)\mu(N-n) \;=\; o(N).$$

This paper establishes what is exactly computable about that scalar. Its second moment has a closed form, $V(N) = \sum_{v<N} \mu^2(v)\Lambda(N-v)^2 \sim \mathfrak{A}(N) N \log N$, whose local factor is $\mathfrak{A}(N) = \prod_{q\nmid N}(1-1/(q(q-1)))$ and *not* the Hardy–Littlewood singular series (Proposition 1); the same constant is, not coincidentally, the deficiency in Huang–Li's own lower bound. The aggregate second moment of the truncated convolution is an exact unconditional identity in terms of the prime-pair sum and the binary Chowla sum (Lemma 5), the truncation being load-bearing.

We then quantify why the natural conditional route does not close. Writing $\rho(N) = \operatorname{Var} C(N)/V(N)$, the coefficient amplifying the Chowla correlation $S(h)$ grows like $N/(\mathfrak{A}(N)\log N)$ and is nonnegative, so a hypothesis $|S(h)| \le \varepsilon$ gives $\rho \to 1$ only for $\varepsilon = o(\log N/N)$ — a factor $N/\log N$ beyond Chowla — and no such bound is true, because the measured absolute budget already exceeds 1 by an order of magnitude and grows (Proposition 7). The only surviving route is signed cancellation across h, and we state that as an open question.

Finally we delimit what can be measured about $C(N)$ at all. Replacing μ by an arbitrary sign pattern on its own support leaves $V(N)$ unchanged, so any estimator reproduced under that replacement is not measuring μ (Lemma 9); in particular no centred estimator calibrates ρ. The obstruction is not information-theoretic — estimators reading multiplicativity separate μ from a coin cheaply, and the field determines μ by an invertible additive convolution (Proposition 11) — but the constructive inverse is numerically dead, its filter growing like 1.92^m.

Net progress toward Goldbach: zero. What is offered is the exact facts, and the precise reason two natural routes to $C(N) = o(N)$ are not routes.

1 Introduction

1.1 The reduction

Let N be a large even integer, Λ the von Mangoldt function, μ the Möbius function, $\tilde{r}(N) = \sum_{n<N} \Lambda(n)\Lambda(N-n)$, and

$$\mathfrak{A}(N) = \prod_{q\nmid N}\Big(1 - \frac{1}{q(q-1)}\Big), \qquad \mathfrak{S}(N) = 2\prod_{p>2}\Big(1 - \tfrac{1}{(p-1)^2}\Big)\prod_{\substack{p\mid N\\ p>2}}\Big(1 + \tfrac{1}{p-2}\Big), \tag{1}$$

q and p denoting primes. Huang and Li [1] proved that binary Goldbach for all sufficiently large even N follows from $EH_\mu(N^{\theta'})$ for a single $\theta' > 1/2$. The companion paper [4] evaluates that hypothesis: one half of it is an unconditional theorem, and the other half is, by an unconditional identity, equivalent to Huang–Li's equation (22) — which yields the asymptotic $\tilde{r}(N) \sim \mathfrak{S}(N)N$ exactly when

$$C(N) \;=\; \sum_{n<N} \Lambda(n)\mu(N-n) \;=\; o(N). \tag{2}$$

This paper is about (2).

The constant $\mathfrak{A}(N)$ is Huang–Li's equation (7), and it enters their argument through the triangle inequality $|C(N)| \le \sum_{n<N} \Lambda(n)\mu^2(N-n)$, whose right-hand side counts squarefree shifted primes. Proposition 1 evaluates exactly that count. So the deficiency $1-\mathfrak{A}(N)$ in their lower bound is one minus the local density of the second moment of C, and the two sides of the problem meet at a constant neither names.

1.2 Results

Section 2 gives the exact facts: Proposition 1 (the scale), Lemma 5 (the exact aggregate identity). Section 3 gives the obstruction to the conditional route (Proposition 7). Section 4 gives the two controls that delimit measurement (Lemma 9, Proposition 11). Section 5 records the size of the margin the problem demands.

1.3 What is not claimed

- **No law for $C(N)$ is conjectured here.** In particular no distributional statement of the form $C(N) = m(N) + \sqrt{V(N)}\, G(N)$ is asserted.
- **No level or rate for $\rho = \operatorname{Var} C/V$ is quoted.** By Lemma 9 the centred estimator cannot distinguish μ from a sign pattern, so any such figure calibrates nothing.
- **The decoupling is not assumed.** Passing from $\operatorname{OffDiag}(N)$ to $\sum_{h\neq 0} c(h)S(h)$ replaces values sampled on a sparse set of primes by an average over all integers; that step is not algebra and is not assumed anywhere below. Proposition 7 shows the conditional route fails *even granting it.*
- **No progress toward Goldbach.** Nothing here bounds $C(N)$.

2 The wall, exactly

2.1 The exact scale

Proposition 1 (the exact scale)**.** *Let*

$$V(N) = \sum_{v<N} \mu^2(v)\,\Lambda(N-v)^2, \qquad W(N) = \sum_{w<N} \Lambda(w)^2.$$

Then $V(N) = W(N)\,\mathfrak{A}(N)\,(1+o(1))$, and consequently $V(N) \sim \mathfrak{A}(N)\,N\log N$.

Proof. $\Lambda(w)^2$ is supported on prime powers with weight $(\log p)^2$, so with $w = N - v$,

$$V(N) = \sum_{p^k<N} (\log p)^2 \mu^2(N-p^k) = \sum_{p<N} (\log p)^2 \mu^2(N-p) + O(\sqrt{N}\log^2 N),$$

a $(\log p)^2$-weighted count of *squarefree shifted primes.* Its local density at a prime q is 1 when $q \mid N$ — there $q^2 \mid N-p$ forces $q \mid p$, hence $p = q$, a single term — and $1 - 1/\varphi(q^2) = 1 - 1/(q(q-1))$ when $q \nmid N$, since the bad class $p \equiv N \pmod{q^2}$ is then a unit class. The count is then Mirsky's theorem on squarefree values of shifted primes [2], and partial summation converts it to the stated asymptotic. $\square$

Note 2 (the local factor is $\mathfrak{A}$, not $\mathfrak{S}$). The singular series $\mathfrak{S}(N)$ is Hardy–Littlewood's and is correct for the Goldbach *count*; the right object for the second moment of $C(N)$ is $\mathfrak{A}(N)$. Both depend on N only through the set of primes dividing N — though in opposite ways, $\mathfrak{S}$ through a product *over* those primes and $\mathfrak{A}$ through a product over the primes *not* among them — which is why the substitution is easy to make and hard to see. One reconstruction error of exactly this kind divides by $W(N)$ where the definition calls for $V(N)$; the two differ by exactly $\mathfrak{A}(N)$.

Measurement 3 (the local factor, discriminated)**.** The statistic, stated: for a candidate $c \in \{\mathfrak{A}, \mathfrak{S}\}$ put $z_c(N) = \big(V(N)/W(N)\big)/c(N)$; the figure quoted is $\mathrm{sd}\big(z_c/\overline{z_c}\big)$, i.e. the candidate is rescaled to the measured mean so that only its shape in N is judged. Over even $N \in [10^5, 1.6 \cdot 10^7]$ it is 0.000323 for $\mathfrak{A}$ against 0.245235 for $\mathfrak{S}$, a factor of 759.3. The reading has to be stated: subtracting a least-mean multiple rather than dividing gives 0.000317 and 0.299040 instead. The lower cutoff is not cosmetic — the figure for $\mathfrak{A}$ rises as the cutoff falls, reading 0.000346, 0.000398, 0.000470, 0.000529 at cutoffs $5 \cdot 10^4$, 10^4, 10^3, 10^2, while the figure for $\mathfrak{S}$ is unaffected, being dominated by $\mathfrak{S}$'s wrong shape rather than by noise.

Dividing through by $W(N)$ removes the analytic factor exactly, so no asymptotic for $\sum(\log p)^2$ is assumed: the ratio $V(N)/W(N)$ against $\mathfrak{A}(N)$ has mean 1.0000002 with standard deviation 0.0001659 in the top octave, and at $N = 4 \cdot 10^6$ the ratio W/V is 1.270800 against the predicted $1/\mathfrak{A}(N) = 1.270204$. By cells, with cell j the even N divisible by 2 and by the first $j-1$ of $3, 5, 7, 11, 13$:

modulus	2	6	30	210	2310	30030
count	7950000	2650000	530000	75714	6883	529
$V/(W\mathfrak{A})$	1.000000475	1.000000147	1.000000010	0.999999570	0.999999037	0.999999646

(`lab_second_moment.py`).

Measurement 4 (the second-order form)**.** The asymptotic $V \sim \mathfrak{A}N \log N$ converges slowly and should be quoted in its second-order form. The measured ratio $V/(\mathfrak{A}N \log N)$ is 0.927137, 0.933409, 0.939701 at $N = 10^6$, $4 \cdot 10^6$, $1.6 \cdot 10^7$. Partial summation from $\theta(x) \sim x$ gives $\sum_{p \le x}(\log p)^2 \sim x \log x - x$, so the second-order form is $\mathfrak{A}(N)\,(N \log N - N)$, a factor $1 - 1/\log N$ smaller — 0.939716 at the top, against the measured 0.939701. Against that form the ratio reads 0.999482, 0.999134, 0.999984 at the same three N (`lab_second_moment.py`).

2.2 The aggregate second moment, exactly

Lemma 5 (the second-moment identity)**.** *Fix X and let*

$$\widehat{C}(N) \;=\; \sum_{\substack{n+v=N \\ n \le X,\; v \le X}} \Lambda(n)\,\mu(v)$$

be the truncated convolution, whose support runs to $2X$. Then

$$\sum_{N \le 2X} \widehat{C}(N)^2 \;=\; \sum_{|h|<X} M(h)\,P(h),$$

where

$$M(h) = \sum_{v,\, v+h \le X} \mu(v)\mu(v+h), \qquad P(h) = \sum_{w,\, w+h \le X} \Lambda(w)\Lambda(w+h).$$

Proof. Expanding the square as a double sum over (n, v) and (n', v') with $n + v = n' + v'$ and summing over *all* N leaves the single constraint $n - n' = v' - v =: h$, with (n, n') and (v, v') otherwise ranging independently over $[1, X]^2$. Collecting the Λ-pairs and the μ-pairs separately gives the two factors. $\square$

Note 6 (the truncation is load-bearing). The truncation on the left is necessary and it is easy to lose. If one writes instead $\sum_{N\le X} C(N)^2$ with C the untruncated convolution, the three surviving indices (n, v, h) are constrained by $n+v \le X$ — a *simplex* — while $M(h)P(h)$ ranges them over the *box* $[1, X]^2$ and so also counts every pair with $X < n+v \le 2X$. The two sides then differ by a factor near 1.57 at X from 800 to 3200, and the ratio does not tend to 1. The form of Lemma 5 is exact to machine precision at every X tested, to $X = 1.6 \cdot 10^7$.

The identity is exact and unconditional, and it makes the aggregate size of the wall a statement about two shifted correlations, one of which (P) is the Hardy–Littlewood prime-pair sum and the other of which (M) is the binary Chowla sum.

3 What the excess is, and why Chowla does not control it

Write $\rho(N) = \operatorname{Var} C(N)/V(N)$, which is 1 if the $\mu(v)$ on the surviving support are replaced by independent signs. Pointwise, $C(N)^2 = V(N) + \operatorname{OffDiag}(N)$ with

$$\operatorname{OffDiag}(N) = \sum_{h\neq 0} \sum_{p,\,p+h \text{ prime}} (\log p)(\log(p+h))\,\mu(N-p)\mu(N-p-h), \tag{3}$$

which is algebra. Passing from there to $\sum_{h\neq 0} c(h)S(h)$ is *not* algebra: it replaces the values $\mu(N-p)\mu(N-p-h)$, sampled on the sparse set of p with p and $p+h$ both prime, by an average over all u. We do not assume that decoupling. What we record is that even granting it, the conditional route does not close.

Throughout this section

$$c(h) \;=\; \sum_{\substack{p'-p=h\\ p,p'\le X}} (\log p)(\log p'), \tag{4}$$

the sum being over *primes*, not prime powers. This fixes the meaning of c once; the alternative $c(h) = \sum_n \Lambda(n)\Lambda(n+h)$ differs by $1 + O(N^{-1/2})$, since $\psi(x) - \theta(x) \asymp \sqrt{x}$ and the relevant quantity is squared, and nothing below turns on the choice.

Proposition 7 (the amplification). *Let $\Gamma(N) = \big(\sum_{h\neq 0} c(h)\big)/V(N)$ with c as in (4) and $X = N$. Then*

$$\Gamma(N) \;\sim\; \frac{N}{\mathfrak{A}(N)\log N},$$

and consequently a hypothesis $|S(h)| \le \varepsilon$ yields nothing better than $|\rho - 1| \le \varepsilon\,\Gamma(N)$. Because $c(h) \ge 0$ that inequality is sharp, so such a hypothesis can give $\rho \to 1$ only for $\varepsilon = o(1/\Gamma(N)) = o(\log N/N)$.

Proof. The numerator needs no correlation input: summing $c(h)$ over all $h \neq 0$ is exactly $\theta(N)^2 - \sum_{p<N}(\log p)^2$, which is $N^2(1+o(1))$ by the prime number theorem. The denominator is $\mathfrak{A}(N)N\log N(1+o(1))$ by Proposition 1. Dividing gives the asymptotic. The inequality $|\rho - 1| \le \varepsilon\Gamma$ is the triangle inequality applied to $\sum_{h\neq 0} c(h)S(h)$ after division by V, and it is sharp because every $c(h)$ is nonnegative, so no cancellation among the coefficients is available to a hypothesis that controls only $|S(h)|$. ☐

Two consequences. First, Chowla's conjecture asserts $S(h) = o(1)$ for fixed h — and the strongest unconditional results in that direction, the logarithmically averaged two-point theorem of [5] and the shift-averaged bound of [3], give smallness of exactly that order. The strength the display needs is $S(h) = o(\log N/N)$, so the gap is a factor $N/\log N$. Second — and this is what closes the absolute-value route rather than merely costing it a power of log — the true S overspends that budget already. Define the *absolute budget*, which is exactly what the triangle inequality spends,

$$\mathcal{B}(X) = \frac{1}{V(X)} \sum_{0<|h|<X} c(h)\,|S(h)|, \qquad S(h) = \frac{1}{X-|h|} \sum_{|h|<u\le X} \mu(u)\,\mu(u-|h|). \tag{5}$$

Measurement 8 (the budget is already overspent)**.** Measured, $\mathcal{B}(X) = 13.3,\ 17.3,\ 23.1,\ 30.5$ at $X = 2\cdot 10^4,\ 4\cdot 10^4,\ 8\cdot 10^4,\ 1.6\cdot 10^5$: above 1 by an order of magnitude, and growing. Normalising S by X rather than by the number $X - |h|$ of terms gives 7.9, 10.3, 13.8, 18.2 — smaller, same conclusion. Both are reported because the phrase "binary Chowla sum" does not fix the normalisation, and a budget is not interpretable until it does. The amplification itself measures $\Gamma = 1.5128\cdot 10^3,\ 1.8412\cdot 10^4,\ 3.5759\cdot 10^5$ at $N = 10^4,\ 1.6\cdot 10^5,\ 4\cdot 10^6$, with $\Gamma \log N/N = 1.3933,\ 1.3789,\ 1.3590$ against the predicted $1/\mathfrak{A}(N) = 1.270204$ (`audit_amplification.py`).

Hence no bound on $|S(h)|$ alone can give $\rho \to 1$. The only route is *signed cancellation across* h, which is a different object from Chowla's smallness and from the averaged absolute bound of [3]. Naming what supplies that cancellation is open, and we state it as such.

4 Two controls, and what they delimit

Lemma 9 (the coin control)**.** *Let $\varepsilon(v) = \pm 1$ be arbitrary signs on $\{v : \mu(v) \neq 0\}$ and zero elsewhere. Then $\varepsilon^2 = \mu^2$ pointwise, so $V(N)$ is unchanged, and the field $C_\varepsilon(N) = \sum_v \varepsilon(v)\Lambda(N-v)$ has the same exact second moment as $C(N)$ for every N. Consequently any estimator whose output is reproduced when μ is replaced by ε is not measuring μ.*

Proof. Immediate from $\varepsilon^2 = \mu^2$ and the definition of V. □

Note 10 (what the lemma licenses, and what it does not). The implication runs one way: *reproduced under $\varepsilon \Rightarrow$ not measuring μ*. The converse does not follow and is not proved anywhere here, so a statistic that a sign ensemble fails to reach has not thereby been shown to measure μ; it has been shown not to be killed by this particular control. The distinction is quantitatively large. An i.i.d. sign ensemble is *narrower* than a random multiplicative one wherever the object is multiplicative — resolved to 512 draws on one statistic, the i.i.d. spread was 0.6455 of the multiplicative (`audit_cn_coin_deep.py`) — so significance read against an i.i.d. ensemble is larger than it should be. Because narrowness is conservative for killing, every claim this lemma kills stays killed; it is the other direction that fails. An estimator that escapes an i.i.d. ensemble has cleared the weakest available control, and until it has been put against an ensemble sharing μ's multiplicativity, the escape is a fact about the control.

Lemma 9 invalidates the natural measurement of ρ: the centred estimator returns the same value on ε as on μ, so a quoted level for ρ calibrates nothing. *Any claim about* $\operatorname{Var} C$ *relative to V must first exhibit an estimator that distinguishes μ from a sign pattern, and show that it does.*

Proposition 11 (the coin obstruction is arithmetic, not informational)**.** *(i) Estimators separating μ from a sign pattern exist and are cheap, provided they read* multiplicativity *rather than variability. Since $\mu(2v) = -\mu(v)$ for odd v is an exact identity while $\varepsilon(2v)$ and $\varepsilon(v)$ are independent signs,*

$$T(x) \;=\; \frac{1}{x}\sum_{v\le x}\mu(v)\mu(2v) \;\longrightarrow\; -\frac{4}{\pi^2},$$

minus the density of odd squarefree integers, while its analogue for independent signs is $O(x^{-1/2})$.

(ii) The field determines μ. Writing $M = N-2$ and $\widehat{\Lambda}(m) = \Lambda(m+2)$, one has $C(N) = (\widehat{\Lambda} * \mu)(M)$ as an additive convolution with $\widehat{\Lambda}(0) = \Lambda(2) = \log 2 \neq 0$, so $\widehat{\Lambda}$ is invertible and $\mu(M) = \sum_{m<M} a(m)\, C(M-m+2)$ for the inverse filter a.

Hence Lemma 9 is not an information-theoretic obstruction: two different sign patterns give two different fields.

Proof. (i) On odd squarefree v, $\mu(2v) = \mu(2)\mu(v) = -\mu(v)$, so $\mu(v)\mu(2v) = -\mu^2(v)$ there; on even v, $\mu(2v) = 0$. Summing gives $-\#\{v \le x : v \text{ odd squarefree}\} = -(4/\pi^2)x(1 + o(1))$. For independent signs the summands are independent mean-zero, whence the $O(x^{-1/2})$. (ii) An additive convolution by a sequence with nonzero value at 0 is invertible over $\mathbb{Q}$ by forward substitution, which defines a uniquely and gives the stated inversion. □

Measurement 12 (the discriminator, and the price of inverting)**.** At $x = 10^4, 10^5, 10^6, 2 \cdot 10^6$, against $-4/\pi^2 = -0.405285$,

$$T(x) \;=\; -0.405600,\; -0.405270,\; -0.405286,\; -0.405295,$$

while the worst of twenty independent-sign draws on the same support is

$$0.020600,\; 0.004570,\; 0.001214,\; 0.000935.$$

At the top the measured value is a factor 434 above the worst of those twenty draws. That factor is not a number of standard deviations: no standard deviation of the sign ensemble is computed anywhere here, and the worst of twenty draws from a half-normal is about 2σ, so the separation in units of σ is larger than 434 rather than smaller. The factor is quoted because it is what was measured.

The constructive route of (ii) is killed by amplification, not by information. The inverse filter grows geometrically: $|a(m)|$ runs 1.443, 2.287, 2.182, 20.91, 512.3, $3.443 \cdot 10^5$, $1.073 \cdot 10^{14}$, $1.535 \cdot 10^{28}$, $3.146 \cdot 10^{56}$ at $m = 0, 1, 2, 5, 10, 20, 50, 100, 200$, a growth of about 1.92^m. Any numerical use of the inversion is therefore dead at once (`lab_coin_discriminator.py`).

5 The margin the problem demands

The requirement $C(N) = o(N)$ constrains every N, so the figure to quote is the one at the extreme rather than at a typical N. At the extreme a size has to be borrowed rather than measured, and the loan is worth naming. *Suppose* $C(N) = \sqrt{V(N)}\,G(N)$ with G behaving across N like independent standard Gaussians. That is precisely the distributional statement §1.3 declines to assert; it is admitted here, in this section only, and nothing in this paper rests on what follows. Then $\max|C| \approx a_n\sqrt{V(N)}$ with a_n the Gumbel location for the number of even N below the point, and

$$\frac{N}{\max_{N \le X} |C(N)|} \;\approx\; \frac{\sqrt{N}}{a_n\sqrt{\mathfrak{A}\log N}},$$

which is $10^{4.466}$ at $N = 10^{12}$ and $10^{22.842}$ at $N = 10^{50}$ with $a_n = \sqrt{2\log(N/2)}$ and $\mathfrak{A} = 0.787275$. That is the value of (1) on the family the sweep uses — N whose only odd prime factor is 5 — and not a generic one: $\mathfrak{A}$ depends on the prime support of N, and an even N not divisible by 5 carries the extra local factor $1 - \frac{1}{5 \cdot 4}$, so it gives 0.95 times as much (`audit_margin.py`, `audit_constants.py`). Those two N lie five and thirty-eight orders of magnitude beyond anything measured here, and the digits are the arithmetic of the display under the supposition, not a measurement. Under it the quantity is expected to be small by a wide margin and is provably small by none: the gap between what is expected and what is reachable is the whole problem.

6 Open questions

1. $C(N) = o(N)$ itself.
2. Name the mechanism supplying *signed* cancellation across h in (3). Proposition 7 shows that no bound on $|S(h)|$ can substitute for it, and nothing in the literature supplies it.

3. Exhibit an estimator of ρ that provably distinguishes μ from a sign pattern in the sense of Lemma 9, and calibrate it against a random multiplicative ensemble rather than an i.i.d. one (Note 10).

7 Summary

- Proposition 1: $V(N) \sim \mathfrak{A}(N)N \log N$, with the local factor $\mathfrak{A}$ and not the singular series $\mathfrak{S}$; the same constant is the deficiency in Huang–Li's lower bound.
- Lemma 5: the aggregate second moment of the truncated convolution is exactly $\sum_h M(h)P(h)$, unconditionally. The truncation is load-bearing.
- Proposition 7: the coefficient amplifying $S(h)$ grows like $N/(\mathfrak{A} \log N)$ and is nonnegative, and the measured absolute budget already exceeds 1 by an order of magnitude and grows. So no bound on $|S(h)|$ gives $\operatorname{Var} C \sim V$; only signed cancellation across h would.
- Lemma 9 and Proposition 11: no centred estimator calibrates ρ; discriminating estimators exist and must read multiplicativity; the constructive inversion is numerically dead.
- Net progress toward Goldbach: zero.

References

[1] J.-J. Huang and H. Li, *On the connection between the Goldbach conjecture and the Elliott–Halberstam conjecture*, arXiv:2005.03811 [math.NT]; v1 (May 2020), v2 (August 2022). Also published in a Springer volume, `doi:10.1007/978-3-030-67996-5_17`. References here are to the arXiv version, which we have read, and not to the published version, which we have not.

[2] L. Mirsky, *The number of representations of an integer as the sum of a prime and a k-th power free integer*, Amer. Math. Monthly **56** (1949), 17–19.

[3] K. Matomäki, M. Radziwiłł and T. Tao, *An averaged form of Chowla's conjecture*, Algebra Number Theory **9** (2015), 2167–2196.

[4] *An unconditional bound for a Möbius-weighted correlation sum in fixed residue classes, with two consequences for the Huang–Li reduction of Goldbach to Elliott–Halberstam*, companion paper.

[5] T. Tao, *The logarithmically averaged Chowla and Elliott conjectures for two-point correlations*, Forum Math. Pi **4** (2016), e8.

A Reproducibility

Script	Result file	Supports
`lab_second_moment.py`	`lab_second_moment.txt`	Measurements 3, 4
`audit_amplification.py`	`audit_amplification.txt`	Measurement 8
`lab_coin_discriminator.py`	`lab_coin_discriminator.txt`	Measurement 12
`audit_cn_coin_deep.py`	`audit_cn_coin_deep.txt`	the 0.6455 of §4
`audit_margin.py`	`audit_margin.txt`	the extrapolation of §5
`audit_constants.py`	`audit_constants.txt`	$\mathfrak{A} = 0.787275$, §5

Every measurement above states the range on which it was taken, and the statistic and field it was taken of. No computation is used as a premise of Proposition 1, Lemma 5, Proposition 7 or Proposition 11, each of which is proved.